Zero Worlds: The Observer, the Coincidence, an the Collapse

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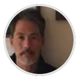

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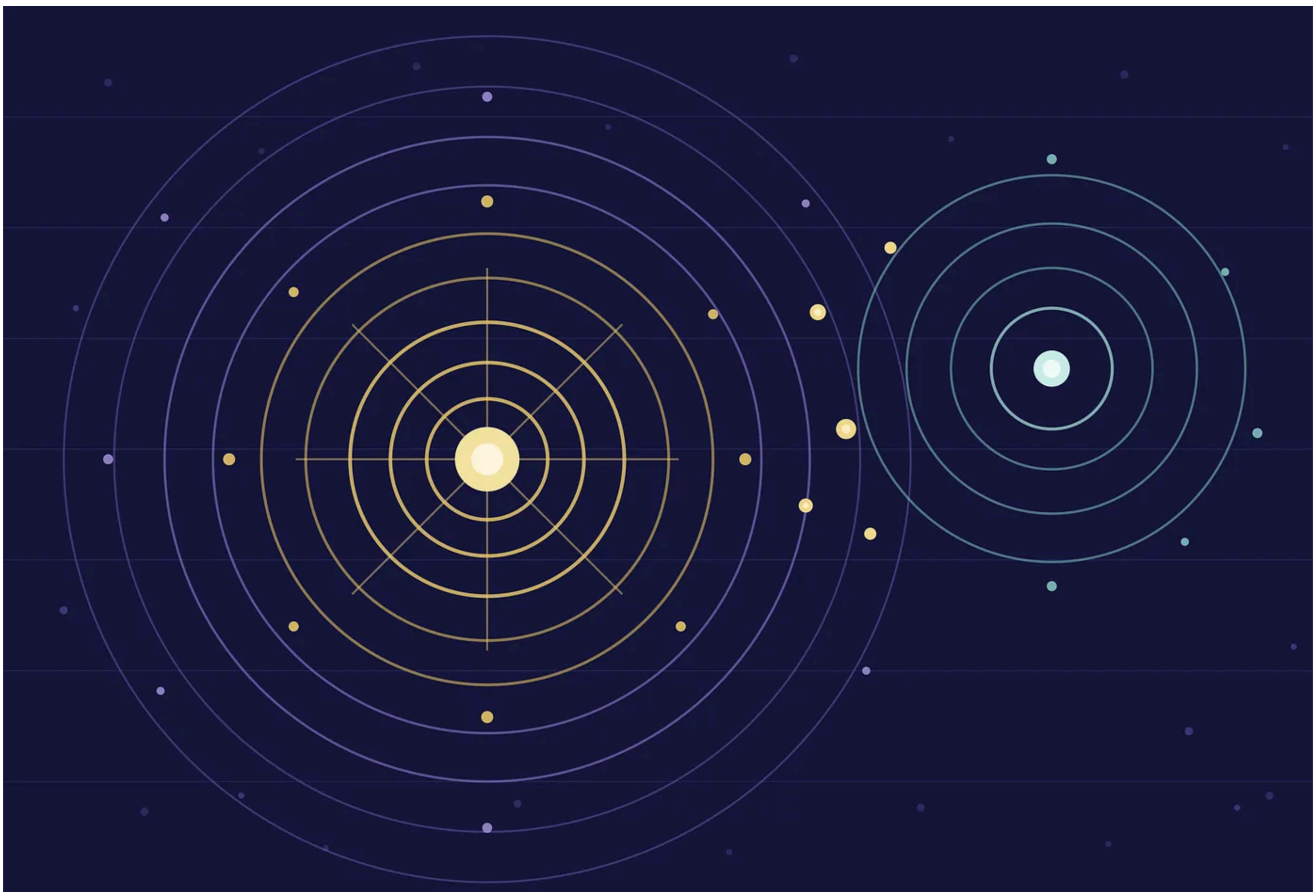

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I. A Physics That Begins Nowhere

Start with a heresy. Suppose there is no external world — not one world, not the branching many worlds of Everett, but *zero* worlds. Suppose the only thing you are entitled to assume is that you are an observer, that you have some finite bundle of d and that in a moment you will have some other bundle of data. What physics can yo build from that?

This is the wager of Markus P. Müller, group leader at the Institute for Quantum Optics and Quantum Information in Vienna and a fellow at the Perimeter Institute his 2020 *Quantum* paper "Law without Law" and a trio of more recent "Algorithmic Idealism" papers, Müller develops what an interviewer once got him to call, smiling "zero-worlds" theory — one that "gives you idealism for free." It is not mysticism. I algorithmic information theory taken with unusual seriousness.

"Law without Law" framed the whole edifice as resting on a *single* postulate: that universal induction determines the chances of what any observer sees next. The late papers unfold this into two. The first (**Self States**): there is a set of "self-states" — finite informational patterns encoding everything true of an observer at a given moment, including all memory. An agent is not fundamentally embedded in spacetime; it *is* its self-state, a standalone pattern. The second (**State Change**): giver that you are in self-state x now, the chance of what you experience *next* is governed universal induction. There is no world underneath doing the deciding. There is only the question that titles Müller's 2024 paper: **what should you believe to experience next?**

II. Solomonoff, Kolmogorov, and the Universal Pric

To make this precise you need two objects from algorithmic information theory.

The **Kolmogorov complexity** *K(x)* of a string x is the length of the shortest program that outputs x on a fixed universal (prefix-free) Turing machine. It is the size of x compressed as far as it can go — a formal measure of simplicity. The **conditional** complexity *K(x|y)* is the shortest program that outputs *x* when *y* is already given for free.

The **universal a priori probability** is Solomonoff's prior:

$$M(x)=\sum_{p:\,U(p)=x*} 2^{-|p|},$$

the probability that a universal monotone machine fed random coin-flips prints something beginning with *x*. Short-program strings dominate the sum, so *M* is a mathematically exact rendering of Occam's razor: simple strings are likely, random strings vanishingly rare. Levin's coding theorem ties the two together: for the discr prefix version, $-\log M(x) = K(x) + O(1)$.

Müller's law of motion is then conditional algorithmic probability:

$$P_{1st}(y|x)\approx \frac{M(xy)}{M(x)}\approx 2^{-K(y|x)}.$$

In Müller's own gloss (with Caroline Jones, in the 2024 Wigner's-friend paper), "*M*(y is large iff *y* is strongly compressible, given *x*." Your next experience is likely to the extent that it is *simple given everything you already are*. Their canonical illustration: if 1^n (a run of ones), then $M(1^m|x) \approx 1$ — induction bets the pattern continues — where two equal-length strings *unrelated* to *x* get roughly equal probability, recovering a principle of indifference.

Two features deserve emphasis. First, the whole apparatus is **uncomputable**: Solomonoff's *M* is only lower semicomputable, because *K* is uncomputable. The ide inductor is a mathematical God, not an algorithm you can run. Second — and this i the payoff — from this solipsistic seed an *external world provably emerges*. Because lo complexity continuations dominate, observers should expect their data to look as if generated by simple, computable, probabilistic laws. Objective reality is not assum it is a theorem, holding asymptotically and approximately.

Müller calls the mechanism that stabilizes this the **Principle of Persistent Regularities**: regularities in your current self-state tend to persist, because inductic bets on them persisting. This is what dissolves the **Boltzmann brain** paradox. A Boltzmann brain's next moment is thermal noise — data *algorithmically uncorrelated* with its rich memories — so $M(y|x)$ for "dissolve into chaos" is tiny, while $M(y|x)$ for "business as usual on Earth" is large. You should never bet on being a Boltzmann brain, however many there are, because the *transition* out of one is algorithmically disfavored. The same first-person move dissolves Wigner's-friend puzzles (there ne be no shared world for two observers to disagree) and reframes the simulation hypothesis (shutting off a simulation need not end its inhabitants, since self-states substrate-independent patterns). An observer is a pattern; the medium is irrelevant

III. The Self in the Conditional: Synchronicity as Mutual Information

Here begins the synthesis — and a caveat that must be stated at the outset. **Müller never mentions Jung, synchronicity, or parapsychology.** What follows is *my* construction: an argument that *if* meaningful coincidence is real in roughly the sen one theorist has proposed, *then* algorithmic idealism already has a natural home for This is a conditional, not a claim of Müller's.

The theorist is Walter von Lucadou, a German physicist-parapsychologist whose **Model of Pragmatic Information** (MPI) reframes Jungian synchronicity — meanin coincidence — not as a signal passing between mind and matter but as an *entanglement-like correlation* within an "organizationally closed" psychophysical syst MPI sits inside Generalized Quantum Theory (Atmanspacher, Römer, Walach), whi abstracts complementarity and entanglement away from microphysics. Its load-bearing axiom is **Non-Transmission** (**NT**): these correlations are real but *cannot car controllable signal*. Pragmatic information itself, following von Weizsäcker, is a prod of two complementary quantities — **novelty** (*Erstmaligkeit*) and **confirmation** (*Bestätigung*) — such that it vanishes if either is zero.

Now the translation. Let *c* be the public context (the shared, describable backgroun *h* the observer's private history (the biography folded into the self-state), and *e* the event that just occurred. Define the coincidence's meaning as the **conditional algorithmic mutual information**:

$$I_K(e:h|c)=K(e|c)-K(e|h,c).$$

In words: how many bits your life saves in describing what just happened. A coincidence is "meaningful" precisely when *e* is hard to compress from public cont alone but easy to compress once your particular history is available — when your biography is the codebook that makes the event short. **Meaning is compression se from a life.**

The crucial question is whether this quantity is *ontic* (it actually shapes the probabi of your next self-state) or merely *epistemic* (a post-hoc gloss). My verdict, sharpened direct textual work on Müller's papers, is strong: **in Müller's framework the ontic reading is already implied — no new postulate is required.** Because $P_{\text{1st}}(y|x) \approx 2^{-K(y|x)}$ and the self-state *x* contains the history *h*, a continuation that shares algorithmic information with your biography has a *smaller conditional complexity* an therefore *already* favored by universal induction. This is not an analogy to the Boltzmann-brain resolution; it is the very same mechanism. Müller uses it to explai why you keep seeing a lawful Earth rather than noise. Pointed at biography rather t physics, it says: continuations that "rhyme" with your particular history are, by the theory's own law of motion, more probable than algorithmically unrelated ones of equal complexity.

So the synthesis does **not** strengthen Müller; it *specializes* him. The bare postulate weights by the total conditional complexity $K(y|x)$ of the whole successor. To isolate $I_K(e:h|c)$ as "the meaning of this coincidence," one adds only a *modeling choice* — a decomposition of the self-state into (c, h, e) and the standard AIT chain rule $K(xy) \doteq K(x) + K(y|x)$ — after which $I_K(e:h|c)$ falls out, up to logarithmic terms, as the boost t $M(xy)/M(x)$ contributed by the match between event and biography. It is implicit in t postulate up to $O(1)$, not bolted on. (One honest wrinkle: Müller notes the current "l model" unrealistically packs the *entire* past into x, an artefact he intends to replace with a model that permits forgetting — under which only the *retained*, salient porti of h would participate, arguably a *more* faithful reading of what "meaningful to me" means.)

From this single identity, Lucadou's entire phenomenological profile follows:

- **Non-signaling.** The effect lives in $K(e|h,c)$ — the term indexed to a *private* histo Promote it to a public signal and you move it into $K(e|c)$: the conditioning on h exactly what you have destroyed. De-index the coincidence and it evaporates. T is the NT axiom, derived rather than postulated.
- **The decline effect.** Controlled replication averages over many observers and histories. But the effect is *the conditional*; averaging over the h's that carry it washes it out. Replication *is* de-indexing. Lucadou observes decline empirically here it is a corollary of where the information lives.
- **Why it can't pick lottery numbers.** Arbitrary tokens — next week's draw — sh essentially zero algorithmic mutual information with any biography. $I\_K \approx 0$, sc there is nothing to boost. The framework predicts meaningful coincidence sho be *useless* exactly where it would be lucrative.
- **Shared synchronicities.** Two people's meaningful coincidence uses a joint hist $H = (h_1, h_2, c)$; Lucadou's "organizationally closed psychophysical system" becon the region over which the conditional is computed.

There is a sharpening the primary sources force. What $M(y|x)$ actually rewards is *lov conditional complexity $K(y|x)$*; this equals *high mutual information* only when the continuation's own complexity $K(y)$ is held fixed. So the honest statement is: *among continuations of comparable intrinsic complexity, the ones that resonate with your history* And one caveat of provenance: the universal machine contributes only an additive constant independent of the data. The machine cannot tell a meaningful coinciden from a meaningless one. **The self lives in the conditional, not the machine.**

The intellectual-honesty ledger: this is a *conditional* synthesis. Synchronicity's empirical reality is contested; Lucadou's own correlation-matrix experiments — wh found large excess correlations early (he reported effects around 5.5 sigma in the 19 and then failed to replicate cleanly — show the very decline that makes them hard t confirm, and later matrix replications returned null. The claim here is architectural not evidential: *if* there is something like MPI to explain, algorithmic idealism expla it for free, through structure it already has.

IV. Penrose–Hameroff: A Kinship That Fails

If MPI slots into Müller like a key into a lock, **Orchestrated Objective Reduction** (Orch OR) does not — and the failure is instructive.

Orch OR, from Roger Penrose and Stuart Hameroff, has three parts. The **Gödelian** (the Penrose–Lucas argument): human mathematicians can see the truth of Gödel sentences their formal systems cannot prove, so mathematical understanding is *non computable*. The **gravitational** part: superposed mass distributions are superposed spacetimes, and this is unstable, collapsing "objectively" in a time $\tau \approx \hbar/E\_G$, where E_G is the gravitational self-energy of the difference (the Diósi–Penrose model). Th **biological** part (Hameroff): these reductions are orchestrated in neuronal microtubules, each collapse a moment of proto-consciousness.

The empirical scoreboard is mixed and worth stating carefully. On one side, the parameter-free Diósi–Penrose collapse model was *experimentally ruled out*: Donadi, Piscicchia, Curceanu, Diósi, Laubenstein and Bassi, reporting in *Nature Physics* (17, 78, 2021), searched for the faint spontaneous radiation the model predicts using a germanium detector 1.4 km underground at Gran Sasso, found none, and concludec that the result "rules out the natural parameter-free version of the Diósi–Penrose model." Max Tegmark's classic 2000 objection (*Physical Review E* 61, 4194) put microtubule decoherence at 10^{-13}–10^{-20} s, "much shorter than the relevant dynamica timescales"; Hagan, Hameroff and Tuszyński rebutted (*Phys. Rev. E* 65, 061901, 2002) that a corrected model "lengthens the decoherence time to 10^{-5}–10^{-4} s." On the othe side, the microtubule *biology* has had genuinely surprising confirmations: room-temperature quantum-optical effects, ultraviolet **superradiance** predicted across mega-networks of more than 10^5 tryptophan transition dipoles (Babcock et al., *J. Ph Chem. B*, 2024), and anesthetics that dampen these effects — while the microtubule-stabilizer epothilone B *delayed* loss of righting reflex in rats by about 69 seconds, a large effect (Cohen's $d \approx 1.9$; eNeuro, 2024). The biology is alive; the specific gravitational-collapse mechanism, in its clean form, is dead.

But the deep issue with Müller is structural, not empirical. **Algorithmic idealism is computable in character and substrate-independent; Orch OR is non-computable and substrate-dependent.** For Müller, a simulated agent experiences reality indistinguishably from a biological one; the pattern is all. For Penrose, consciousne *cannot* be instantiated digitally, because it depends on specific non-algorithmic quantum-gravitational physics that no Turing machine can capture. These are contradictories.

There is a subtlety one must not fudge. Müller's law is *itself* uncomputable — M is c lower semicomputable. So both men invoke a limit of computation. But the *kind* is opposite. Müller's uncomputability is a property of the *ideal inductor*, a prior over computable programs; it is what lets *computable laws emerge* for the observer. Penro non-computability is a property of *individual conscious events* — actual physical dynamics that outrun any algorithm. Müller uses the uncomputable to *explain the computable*; Penrose makes the non-computable *fundamental and local*.

Is any reconciliation possible? One might try to map Penrose's Platonic realm of mathematical truth onto Müller's space of all programs — but the space of progran exactly the computable domain Penrose says the mind transcends, so the identifica defeats itself. One might reinterpret OR events as features of the *emergent* descripti — real to observers, not fundamental — but that demotes precisely what Penrose insists is fundamental. The verdict is honest asymmetry: **Müller and MPI are a nat**

fit; Müller and Orch OR are in genuine tension, despite a superficial kinship. Bot elevate the observer over the derived world. Both stake a claim at the boundary of computation. But one theory dissolves the observer into a pattern that any substrat can carry, and the other anchors it in a biology no algorithm can mimic.

V. Neighbours in the observer-first landscape: Wolfram, Hoffman, and others

Müller's algorithmic idealism did not arrive in a vacuum. It belongs to a loose famil "observer-first" programs that, over the last few decades, have each tried to demote three-dimensional world from *the* fundamental thing to a derived one. The family resemblance is real, but so are the disagreements — and the disagreements are whe the interest lies. The useful way to sort these theories is to ask three questions of ea What is taken as *fundamental*? Does a mind-independent *world* survive underneath t appearances? And is the fundamental level *computable*? Run through those filters, Müller turns out to occupy the most austere corner of the whole landscape.

Wolfram: the same observer, a different ontology

The closest structural cousin is Stephen Wolfram's Physics Project and its central object, the **ruliad** — which Wolfram defines as "the entangled limit of everything th is computationally possible: the result of following all possible computational rules all possible ways." Space, time, particles, and physical law are not built into the rule they emerge from how observers embedded in the ruliad sample it. Crucially, Wolfr argues that the core laws of twentieth-century physics are derivable *not* from the specific rule but from two coarse facts about the observer. In his own words: "if we assume that we are observers who are computationally bounded, and believe we are persistent in time, then we argue that it is inevitable that we must perceive certain laws to be operating — and those laws turn out to be exactly the three central laws twentieth century physics: general relativity, quantum mechanics, and the Second L of thermodynamics."

This is a deep parallel to Müller. Both make the *observer's* characteristics — bounde computation, a drive to compress, a coherent thread of experience — do the work o generating lawful physics, rather than positing the laws directly. Both are, against Penrose–Hameroff, thoroughly *computational*: no non-computable step is needed anywhere. And both predict a broadly objective, lawful world as an emergent, high-probability appearance rather than a starting assumption.

But the ontologies diverge sharply. Wolfram's ruliad genuinely *exists*: it is a unique, "abstractly necessary" computational object, and observers "sample tiny parts of it. That is still, at bottom, a third-person / god's-eye structure — one giant computatic — even though it is observer-sampled. Müller's stance is more radical: there is no underlying world at all, only observer-states (finite bit strings) and induction over them. Objective reality, in his framework, "is not assumed on this approach but rat provably emerges as an asymptotic statistical phenomenon." Call it a zero-world ontology against Wolfram's one-world-that-contains-all-worlds.

There is also a subtler technical contrast worth flagging. Müller's dynamics is weighted: continuations are governed by algorithmic probability, a Solomonoff pric in which shorter programs are exponentially favoured (weight roughly 2^–length). Wolfram's raw ruliad, by contrast, contains all computations *equally*; it carries no explicit simplicity measure. In Wolfram, the pruning that yields recognizable physi comes entirely from the observer's computational boundedness and location in "rul space," not from a prior that privileges simple rules. Whether observer-boundednes does the same work that Solomonoff's prior does in Müller is an open and interesti question — and, tellingly, Müller himself does not engage Wolfram in his core pape the two programs were built independently, and the ruliad–algorithmic-idealism comparison has so far been drawn mainly by third parties rather than by either principal.

Hoffman: consciousness-first versus information-first

Donald Hoffman arrives at a similar destination from evolutionary biology. His **fitness-beats-truth theorem** — a result in evolutionary game theory (Prakash, Stephens, Hoffman, Singh & Fields) — claims that perceptual strategies tuned to fitness generically drive truth-tuned strategies to extinction: natural selection "ofte drives true perceptions to extinction when they compete with perceptions tuned to fitness rather than truth." His **interface theory of perception** then casts spacetime a objects as a species-specific "user interface," like icons on a desktop that hide the machinery. Spacetime, on this view, is "doomed" — a data structure, not fundamen

Hoffman's positive ontology, **conscious realism**, then inverts materialism. As he pu it: "instead of proposing that particles in spacetime are fundamental, and somehow create consciousness when they form neurons and brains, I propose the reverse: consciousness is fundamental, and it creates spacetime and objects." He formalizes this with networks of interacting **conscious agents** whose dynamics are Markovian *Fusions of Consciousness* (2023) he maps those dynamics to "decorated permutations, proposing that "a particle in spacetime is a projection of the Markovian dynamics c communicating class of conscious agents," linking his agents to the amplituhedron and other structures "beyond spacetime."

The parallels to Müller are strong: both deny that the 3D spatiotemporal world is fundamental, and both treat it as a derived, useful, compressed representation — Hoffman's "interface" playing the role of Müller's de-indexed public world. But the primitive is different, and this is the crux. Hoffman makes *phenomenal consciousness* fundamental: his conscious agents have irreducible experiences, and qualia are primitives. Müller is deflationary: observer-states are *informational/computational* states, and the dynamics is universal induction. He needs no irreducible qualia — o algorithmic probability over bit strings. So the axis is sharp: is the primitive *experie* (Hoffman) or *information* (Müller, Wolfram)? The mechanism differs too. Hoffman gives an *evolutionary* argument for why our world-model is a useful fiction; Müller gives an *algorithmic-information* argument (compressibility, Occam) for why a stable objective world emerges. They answer the same question — why does the world loc objective if it isn't fundamental? — with complementary machinery. (Worth noting: Hoffman's conscious-agent networks are themselves described as computationally universal, so, like Müller and Wolfram and unlike Orch OR, his formalism sits on th *computable* side of the ledger even though its primitive is phenomenal.)

The others, briefly, and Müller's distinctive position

Three further reference points sharpen the picture. **QBism** (Fuchs, Mermin, Schack treats the quantum state as "an agent's personal probability assignments, reflecting subjective degrees of belief about the future content of his experience." That is radically first-person, and Müller explicitly positions himself near it — but with a decisive addition: QBism leaves the agent's prior open, whereas Müller supplies a specific universal one. He can be read as QBism with the missing prior filled in by algorithmic probability. **Wheeler** is the shared ancestor: his slogans "it from bit" ar "law without law" name the whole tradition, and Müller borrows the latter as the ti of his foundational paper, quoting Wheeler's view that "when we view each of the la of physics […] as at bottom statistical in character, then we are at last able to forego idea of a law that endures from everlasting to everlasting." **Tegmark**'s Mathematica Universe Hypothesis makes mathematical structure fundamental — "all structures that exist mathematically also exist physically" — but it notoriously lacks a principl measure over its infinity of structures; Müller's algorithmic prior is arguably exactl the measure the MUH is missing, a gap Tegmark himself partly concedes by later restricting to computable structures to "alleviate the cosmological measure problem

Two contrasts round it out. **Integrated Information Theory** (Tononi) is a foil, not a cousin: it starts from a physical world and asks which systems in it are conscious, given a measure Φ of "irreducible, intrinsic cause–effect power." That is substrate-t and third-person — closer in spirit to Orch OR's substrate-dependence than to Müller's substrate-independence. And the **simulation hypothesis** is not so much a rival as a question Müller dissolves: because substrate does not matter and continuation is fixed by algorithmic probability, "shutting down computer simulati does not generally terminate its inhabitants," and the "are we simulated?" question loses its bite.

Seen against this map, algorithmic idealism is the most deflationary of the observer first programs. It keeps no world underneath (unlike Wolfram and Tegmark), no irreducible qualia (unlike Hoffman), and no privileged rule or law (unlike ordinary physics) — only observer-states and a universal prior. Its most intriguing role may b as a *supplier of the missing measure* for its neighbours: the prior QBism leaves open, t measure Tegmark lacks, and perhaps a principled version of the simplicity weightin that Wolfram's boundedness assumption gestures at but never states. That is a genu unifying ambition — though it should be held lightly. These thinkers did not build their systems to be merged, they rarely cite one another, and several of the correspondences here are suggestive analogy rather than proven equivalence.

VI. The Observer, Made Fundamental

What unites these three frameworks is a single reversal. For three centuries physics asked "what is the case in the world?" and derived the observer as an afterthought. Müller, Lucadou, and Penrose each, in incompatible ways, put the observer first — Müller as an information-theoretic self-state, Lucadou as the meaning-generating p of a closed psychophysical system, Penrose as the seat of non-computable insight.

The lesson of the synthesis is that *how* you make the observer fundamental decides everything downstream. Make it a compressible pattern and meaning becomes mut information, coincidence becomes a bias already latent in the law of motion, and consciousness ports freely across substrates. Make it a non-computable event in a specific tissue and you get a mind that no simulation can hold — and a theory at wa with the algorithmic view. The coincidence and the collapse pull in opposite directions. Zero worlds can accommodate the first, elegantly, for free. It cannot, without ceasing to be itself, accommodate the second.

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"Meaning is compression seen from a life" formalizes something I keep circling from a comple
different direction: that a coincidence isn't a message crossing some gap between mind and
it's what happens when a continuation shares algorithmic information with a history that was
actually separate from it. No signal has to cross anything if the biography was already part of
generating the next state.

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